

A PROJECT REPORT

ON

College Placement and Recruitment Management System

Submitted by

25RH1A05NN S.Sreeja
25RH1A05NP S.Rishika
25RH1A05NQ S.Bhavani



Department Of Computer Science And Engineering

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN

(Autonomous Institution-UGC, Govt. of India)

(Affiliated to JNTU, Hyderabad, Approved by AICTE - - ISO 9001:2015 Certified)

Accredited by NAAC with 'A+' Grade Affiliated to JNTUH, Approved by AICTE, ISO 9001:2015

*Certified Institution AAAA+ Rated by Digital Learning Magazine,
AAAARatedbyCareers360Magazine*

Maisammaguda, Dhullapally, Secunderabad, Kompally-500100

AUGUST 2026

MALLA REDDY ENGINEERING COLLEGE FOR WOMEN



(Autonomous Institution-UGC, Govt. of India)

(Affiliated to JNTU, Hyderabad, Approved by AICTE - -
ISO 9001:2015 Certified) Accredited by
NAAC with 'A+' Grade **Maisammaguda,
Dhulapally, Secunderabad -500100.**

CERTIFICATE

This is to certify that this is the bonafide record of the project titled

Blood Bank Donor and Blood Stock Management System

Submitted by

25RH1A05NN S.Sreeja
25RH1A05NP S.Rishika
25RH1A05NQ S.Bhavani

Of third year B.Tech, in the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer science and Engineering during the year **2025-2029**. The results embodied in this **project report** have not been submitted to any other university or institute for the award of any degree or diploma.

INTERNAL GUIDE

Mrs. B.Vasantha

HOD

DR.V.PRADEEP

ACKNOWLEDGEMENT

We feel ourselves honored and privileged to place our warm salutation to our college **Malla Reddy Engineering College for Women** and department of Computer Science and Engineering which gave us the opportunity to have expertise in engineering and profound technical knowledge.

We would like to convey thanks to our internal project guide **Vasantha mam** for her regular guidance and constant encouragement and we are extremely grateful to her for her valuable suggestions and unflinching co-operation throughout project work.

We express our heartiest thanks to **Dr. PRADEEP SIR** Head of the Department, Information Technology for encouraging us in every aspect of our project

We wish to convey gratitude to our Principal **Dr. Y. MADHAVEE LATHA**, for providing us with the environment and means to enrich our skills and motivating us in our endeavor and helping us realize our full potential.

**With Regards and
Gratitude**

**S.Sreeja(23RH1A05NN)
S.Rishika(23RH1A05NP)
S.Bhavani(23RH1A0NQ)**

ABSTRACT

The **Blood Bank Donor and Blood Stock Management System** is a software application designed to manage blood donors, blood groups, and available blood stock efficiently. The system helps blood banks maintain accurate records of donors and monitor the availability of different blood groups.

The system allows staff to register new donors, store donor details, record blood donations, and update blood stock whenever blood is collected or issued. It also helps users quickly search for available blood based on blood group and quantity. This reduces manual paperwork, saves time, and minimizes errors in maintaining records.

The main objective of this system is to provide a simple, organized, and reliable way to manage blood bank activities. By maintaining updated donor and stock information, the system can help blood banks respond more effectively to blood requirements and improve overall management.

The **Blood Bank Donor and Blood Stock Management System** is a computerized system developed to simplify and improve the management of blood bank operations. Managing donor information and blood stock manually can be time-consuming and may lead to errors, duplication of records, or difficulty in finding the required blood group. This system provides a centralized platform for storing, updating, and retrieving important blood bank information.

The system maintains complete records of blood donors, including their personal details, blood group, contact information, and donation history. It also manages blood stock according to different blood groups and keeps track of the available quantity. Whenever blood is donated, the stock can be updated, and when blood is issued, the corresponding quantity can be reduced. This helps staff maintain accurate and up-to-date stock information.

The system provides features such as **donor registration, donor record management, blood stock management, blood search, blood issue records, and report generation**. Authorized users can easily search for a particular blood group and check its availability whenever required. The system can also help identify suitable donors when a specific blood group is needed.

The main objective of this project is to reduce manual work, improve data accuracy, save time, and make blood bank management more efficient. It provides a user-friendly and organized method for handling donor and blood stock records. Overall, the system can support better coordination between donors and blood banks and contribute to faster management of blood availability during emergencies.

INDEX

Title	Page no
Abstract	i
1. Introduction	1
2. Requirements	2
3. Analysis	3
4. Design	4-5
5. Implementation	6-7
6. output screens	8
7. Conclusion& future enhancements	9
8. References	10

1. INTRODUCTION

Blood is one of the most important resources required for saving human lives. Blood is needed during accidents, surgeries, medical treatments, childbirth, and various emergencies. Blood banks play an important role in collecting, storing, and providing blood to patients when required. Therefore, maintaining accurate information about blood donors and available blood stock is very important.

In many blood banks, managing donor details and blood stock using manual records can be difficult and time-consuming. Maintaining large numbers of records on paper may result in errors, duplicate entries, loss of information, and difficulty in finding the required blood group quickly. It can also make it difficult to know the exact quantity of blood available at a particular time.

The **Blood Bank Donor and Blood Stock Management System** is designed to solve these problems by providing a computerized platform for managing blood bank information. The system stores donor details such as name, age, contact information, blood group, and donation history. It also maintains information about the quantity of blood available for different blood groups.

The system allows authorized users to add new donor records, update existing information, record blood donations, manage blood stock, and search for the required blood group. When blood is collected, the stock can be increased, and when blood is issued to a patient, the stock can be updated accordingly. This helps maintain accurate and up-to-date records.

The main purpose of this project is to **reduce manual work, improve accuracy, save time, and make blood bank operations easier to manage**. By providing quick access to donor and blood stock information, the system can help blood bank staff respond more efficiently to blood requirements.

Overall, the Blood Bank Donor and Blood Stock Management System provides a simple, organized, and efficient approach to managing blood donors and blood inventory. It can improve record management and support the effective availability of blood when it is needed.

2. REQUIREMENTS

The **Blood Bank Donor and Blood Stock Management System** requires both hardware and software components to develop and operate the application successfully. The requirements are divided into functional requirements, hardware requirements, software requirements, and user requirements.

1. Functional Requirements

The system should provide the following functions:

- **Donor Registration:** Add and store new donor details such as name, age, gender, blood group, phone number, and address.
- **Donor Management:** View, update, and delete donor information when required.
- **Blood Donation Records:** Record details of blood donations made by donors.
- **Blood Stock Management:** Maintain the quantity of blood available for each blood group.
- **Blood Stock Update:** Increase stock when blood is donated and decrease stock when blood is issued.
- **Blood Search:** Search and check the availability of a particular blood group.
- **Blood Issue Management:** Record blood issued to patients or hospitals.
- **Patient Details:** Store basic information about patients who require blood.
- **Reports:** Generate reports about donors, donations, blood stock, and blood issued.
- **Login and Security:** Provide authorized users with secure access to the system.

2. Hardware Requirements

The basic hardware requirements are:

- Computer or laptop
- Minimum 4 GB RAM
- Minimum 10 GB free storage
- Keyboard and mouse
- Monitor/display
- Internet connection, if the system uses online features

3. Software Requirement

The software requirements may include:

- **Operating System:** Windows/Linux
- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python / Java / PHP
- **Database:** MySQL
- **Development Environment:** VS Code / Eclipse / IntelliJ IDEA
- **Web Browser:** Google Chrome, Microsoft Edge, or Firefox

4. Database Requirements

The database should store information such as:

- Donor details
- Blood group details
- Donation records
- Blood stock information

3.ANALYSIS

The analysis phase of the **Blood Bank Donor and Blood Stock Management System** focuses on understanding the problems in the existing blood bank management process and identifying the requirements of the proposed system. The main purpose is to develop a system that can manage donor information, blood donations, blood stock, and blood requirements in an organized manner.

1. Existing System

In the existing system, blood donor and stock information may be maintained using registers, paper files, or separate computer files. Managing a large number of records manually requires more time and effort. It can also make searching, updating, and maintaining records difficult.

Some common problems are:

- Manual entry of donor information.
- Difficulty in searching for a particular blood group.
- Possibility of incorrect or duplicate records.
- Difficulty in maintaining updated blood stock.
- More paperwork and storage requirements.
- Time-consuming report preparation.
- Difficulty in tracking donation history.

2. Proposed System

The proposed system provides a computerized solution for managing blood bank activities. It stores all important information in a centralized database, allowing authorized users to access and update records easily.

The proposed system includes:

- Donor registration and management.
- Blood group management.
- Blood donation recording.
- Blood stock tracking.
- Patient information management.
- Blood issue records.
- Searching for available blood groups.
- Report generation.
- Secure user login.

3. Feasibility Analysis

Technical Feasibility:

The system can be developed using commonly available technologies such as HTML, CSS, JavaScript, Python/Java/PHP, and MySQL.

Economic Feasibility:

The system reduces paperwork and manual effort. It can be developed using low-cost or open-source software tools.

Operational Feasibility:

The system is designed with a simple interface so that blood bank staff can learn and use it easily.

4.DESIGN

The design phase of the **Blood Bank Donor and Blood Stock Management System** defines the structure, modules, database, and user interface of the proposed system. The main aim of the design is to create a simple, efficient, secure, and user-friendly system for managing donors and blood stock.

1. System Architecture

The system follows a basic three-layer architecture:

User → Application → Database

- **User:** Blood bank staff or authorized administrator.
- **Application:** Handles donor registration, blood stock, donations, and blood issue operations.
- **Database:** Stores all donor, patient, donation, and blood stock information.

2. Main Modules

The system is divided into the following modules:

A. Login Module

- Provides secure access to authorized users.
- Verifies username and password.
- Prevents unauthorized access.

B. Donor Management Module

- Add new donor details.
- View donor information.
- Update donor records.
- Delete records when required.
- Maintain donation history.

C. Blood Donation Module

- Record each blood donation.
- Store donor ID, blood group, donation date, and quantity.
- Automatically update the blood stock.

D. Blood Stock Module

- Maintains the available quantity of different blood groups.
- Displays current stock.
- Updates stock after donation or blood issue.
- Helps identify low-stock blood groups.

E. Patient Management Module

- Stores basic patient details.
- Records required blood group and quantity.
- Maintains blood requirement information.

F. Blood Issue Module

- Records blood issued to patients or hospitals.
- Updates the available blood stock.
- Maintains issue history.

G. Report Module

- Generates donor reports.
- Displays blood stock reports.
- Provides donation and blood issue reports.

3. Database Design

The database can contain the following tables:

Table	Important Fields
Donor	Donor_ID, Name, Age, Gender, Blood_Group, Phone, Address
Donation	Donation_ID, Donor_ID, Blood_Group, Date, Quantity
Blood_Stock	Stock_ID, Blood_Group, Quantity
Patient	Patient_ID, Name, Age, Blood_Group, Contact
Blood_Issue	Issue_ID, Patient_ID, Blood_Group, Quantity, Date

4. Database Relationships

The main relationships are:

- One **Donor** can have multiple **Donation** records.
- Each **Donation** belongs to one donor.
- Each blood group has a corresponding **Blood Stock** record.
- A **Patient** can have one or more blood issue records.
- Each **Blood Issue** updates the corresponding blood stock.

5. User Interface Design

The system can contain the following screens:

1. **Login Page**
2. **Dashboard**
3. **Donor Registration Page**
4. **Donor Details Page**
5. **Blood Donation Page**
6. **Blood Stock Page**
7. **Patient Registration Page**
8. **Blood Issue Page**

The interface should use simple forms, buttons, tables, and search options so that users can easily perform operations.

5.IMPLEMENTATION

The implementation phase of the **Blood Bank Donor and Blood Stock Management System** involves converting the system design into a working application. The system is implemented using a frontend for user interaction, a backend for processing requests, and a database for storing and managing information.

1. Technology Used

The system can be implemented using the following technologies:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python / Java / PHP
- **Database:** MySQL
- **Development Tool:** VS Code
- **Operating System:** Windows or Linux
- **Web Browser:** Google Chrome / Microsoft Edge

2. Login Implementation

A login page is created for authorized users. The user enters a username and password, and the system verifies the entered information with the user database. If the details are correct, the user is directed to the dashboard.

3. Donor Management Implementation

The donor management module allows the staff to enter and maintain donor information. Details such as donor ID, name, age, gender, blood group, phone number, and address are stored in the database.

The module provides options to:

- Add new donors.
- View donor details.
- Update donor information.
- Delete records when necessary.
- Search for donors.

4. Blood Donation Implementation

When a donor gives blood, the donation details are entered into the system. The system records the donor ID, blood group, donation date, and quantity of blood collected. After successfully recording the donation, the corresponding blood stock is updated.

Donation Recorded → Blood Group Identified → Stock Increased

5. Blood Stock Implementation

The blood stock module maintains the available quantity of different blood groups. Whenever blood is donated, the quantity is increased. Whenever blood is issued, the quantity is decreased.

This allows staff to check the current availability of blood quickly.

6. Patient Management Implementation

The patient module stores basic patient information such as patient ID, name, age, required blood group, contact details, and required quantity. This information can be used while processing blood requests.

7. Testing

After implementation, each module is tested to ensure that it works correctly. The following operations are tested:

- User login.
- Donor registration.
- Donor record update.
- Blood donation entry.
- Blood stock update.
- Blood availability search.
- Patient registration.
- Blood issue.
- Report generation.

8. Overall Implementation Flow

Login → Dashboard → Donor Management → Donation → Blood Stock → Patient Request → Blood Availability → Blood Issue → Reports

Conclusion

The implementation of the Blood Bank Donor and Blood Stock Management System provides a computerized method for managing important blood bank activities. It reduces manual work, improves record accuracy, makes blood stock tracking easier, and provides quick access to donor and blood availability information.

6. OUTPUT SCREEN

1. Login Screen

- Username
- Password
- Login button

2. Dashboard Screen

Shows:

- Total Donors
- Total Blood Stock
- Total Donations
- Blood Issued
- Quick navigation buttons

3. Donor Registration Screen

Fields:

- Donor ID
- Donor Name
- Age
- Gender
- Blood Group
- Phone Number
- Address
- **Register** button

4. Donor Details Screen

Displays donor records in a table:

Donor Id	Name	Age	Blood Group	Phone
D001	Rahul	25	O+	9121500234
D002	Priya	28	B+	9224537891

5. Blood Stock Screen

Displays available blood:

Blood Group	Available Units	Status
A+	25	Available
A-	8	Low
B+	30	Available
B-	6	Low
O+	40	Available
O-	5	Low
AB+	15	Available
AB-	4	Low

7. Conclusion

The **Blood Bank Donor and Blood Stock Management System** is an effective computerized solution developed to manage and organize the major activities of a blood bank. The system provides a centralized platform for maintaining donor information, blood donation records, blood stock details, patient information, and blood issue records. It helps replace time-consuming manual processes with a more systematic and efficient approach.

The system makes it easier for blood bank staff to register new donors and maintain their information. Donor details such as name, age, blood group, contact information, and donation history can be stored and retrieved whenever required. This helps maintain proper records and makes it easier to identify donors based on their blood group.

Another important feature of the system is **blood stock management**. The system keeps track of the quantity of different blood groups available in the blood bank. When a donation is recorded, the available stock is updated accordingly. Similarly, when blood is issued to a patient, the stock is reduced. This provides accurate information about the current blood availability and helps staff manage blood inventory effectively.

The system also simplifies the process of handling blood requirements. Staff can search for a particular blood group and check its availability quickly. Patient and blood issue details can also be maintained for future reference. This reduces paperwork, saves time, and improves the overall organization of blood bank records.

The project also provides better **data accuracy, security, accessibility, and reliability** compared with traditional manual record-keeping methods. Proper validation and user authentication can help protect important information and prevent unauthorized access. The implementation of this system shows how information technology can be used to improve the management of healthcare-related services. Although the system is mainly designed for academic and management purposes, its features can be further developed to meet the requirements of real-world blood banks.

In the future, the system can be enhanced by adding **online donor registration, donor appointment scheduling, emergency blood request facilities, automatic notifications, blood donation reminders, hospital connectivity, and advanced report generation**. Integration with mobile applications could also allow donors and authorized staff to access relevant information more conveniently.

In conclusion, the **Blood Bank Donor and Blood Stock Management System** helps create a faster, more organized, and reliable method of managing blood bank information. It reduces manual effort, improves stock monitoring, simplifies donor management, and provides quick access to important records. Therefore, the proposed system can significantly improve the efficiency of blood bank management and support better coordination of blood availability.

8. REFERENCES

1. World Health Organization (WHO), “**Blood Safety and Availability**,” World Health Organization.
2. World Health Organization (WHO), “**Guidance on Centralized and Decentralized Blood Systems**,” WHO.
3. American Red Cross, “**Blood Donation and Blood Services**,” American Red Cross.
4. Indian Red Cross Society, “**Blood Bank and Blood Donation Services**,” Indian Red Cross Society.
5. Elmasri, R. and Navathe, S. B., **Fundamentals of Database Systems**, Pearson Education.
6. Silberschatz, A., Korth, H. F. and Sudarshan, S., **Database System Concepts**, McGraw-Hill.
7. Herbert Schildt, **Java: The Complete Reference**, McGraw-Hill Education.
8. Ramez Elmasri and Shamkant B. Navathe, **Database Management Systems**, Pearson Education.
9. HTML, CSS and JavaScript documentation and learning resources were referred to for designing the user interface and implementing web-based features.
10. MySQL documentation was referred to for database creation, table management, queries, and data storage.
11. Project-related academic materials and classroom notes were referred to for system analysis, design, implementation, and database concepts.

Web References

- **World Health Organization (WHO):** Blood safety and availability information.
- **Indian Red Cross Society:** Information related to blood donation and blood bank services.
- **MySQL Documentation:** Database management and SQL concepts.
- **MDN Web Docs:** HTML, CSS, and JavaScript development concepts.